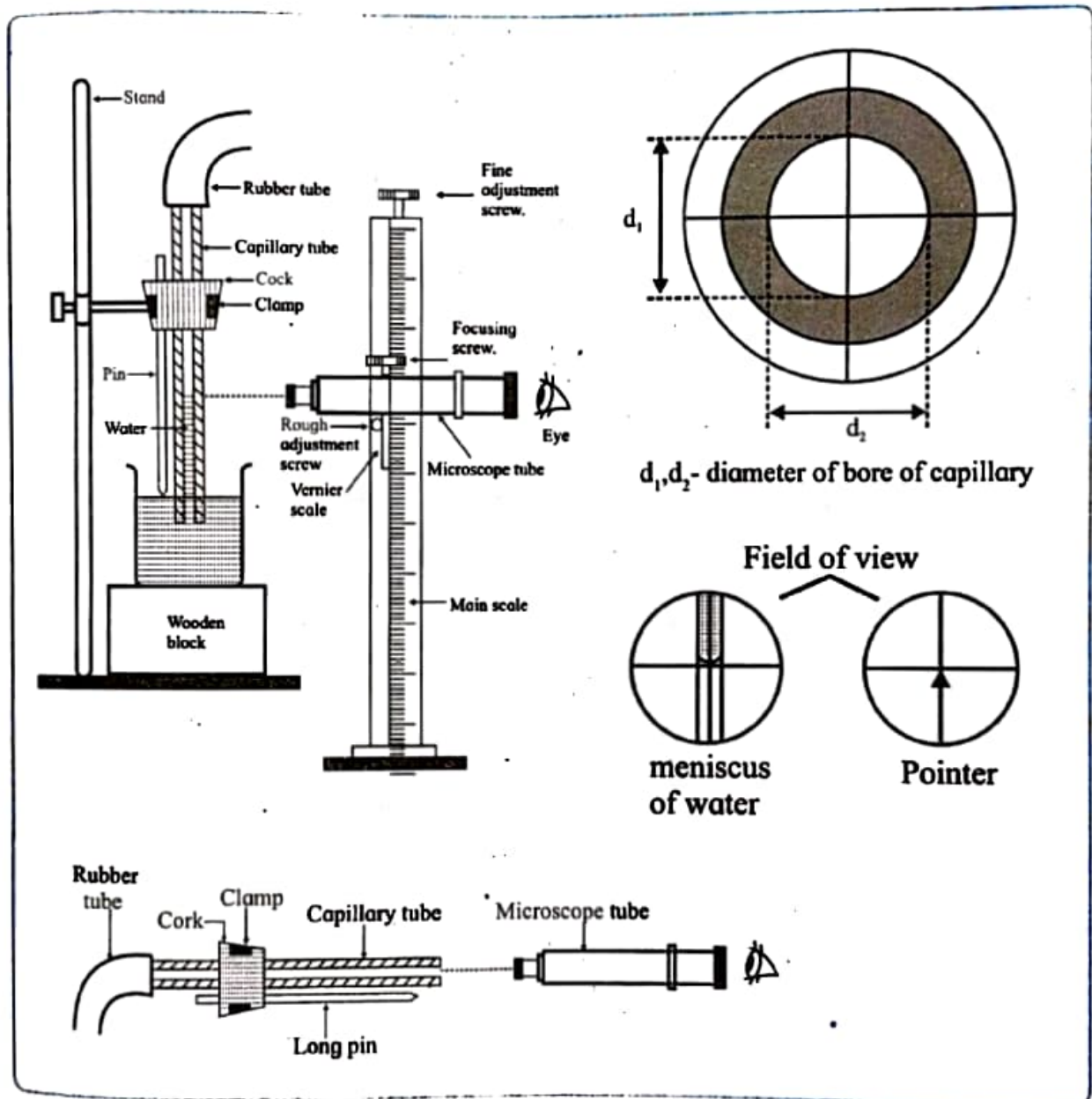


EXPERIMENT NO. 2 SURFACE TENSION

Aim: To determine the surface tension of the given liquid (water) by capillary rise method.

Apparatus : Travelling microscope, beaker, capillary tube, cork with pointer, stand with clamp, water.

Diagram :



Theory:

At N.T.P, when a liquid (water) rises in a Capillary tube the weight of the column of the water of density ρ ($\rho=1$) below the meniscus is supported by the up ward force of surface tension acting around the circumference of the point of contact. Therefore Approximately

$$\text{(for water)} \quad 2\pi rT = \pi r^2 h \rho g \quad \text{or} \quad T = \frac{h \rho g r}{2}$$

Formula :

$$\therefore T = \frac{hrg}{2}$$

where, T = surface tension of liquid

h = height of liquid column in capillary tube

r = inner radius of the cross section of bore of the capillary

ρ = density of the liquid (water)

θ = angle of contact between water and glass

For water, $\rho = 1.0 \text{ g/cm}^3$

Angle of contact between water and glass is very small.

i.e. $\theta = 0^\circ$, $\cos\theta = 1$

Procedure:-**To measure inner diameter of bore (d)**

1. Determine the least count (L. C.) of travelling microscope.
2. Clamp the capillary horizontally with the help of retort stand.
3. Focus the microscope on the bore of the capillary.
4. Adjust the vertical cross wire such that it is tangential to one end of the bore. Record the reading on the horizontal scale (x).
5. With the help of the fine adjustment screw make the vertical cross-wire tangential to the other end of the bore. Record the reading on the horizontal scale (y).
6. Repeat the steps 4 and 5 for horizontal cross wire and record the reading on the vertical scale.
7. Immerse the capillary tube vertically in the beaker containing water. To measure h_1 & h_2
8. Remove the air bubbles by pressing the rubber tube and adjust the level of water in the beaker so that meniscus in the bore of capillary should be above the edge of the beaker and below the lower edge of the capillary stand.
9. Adjust the tip of the pointer so that it just touches the level of water. Focus the travelling microscope on the meniscus such that the horizontal cross -wire is tangential to the convex surface of the meniscus. Record the reading on vertical scale (h_1).
10. Remove the beaker without disturbing the position of the capillary tube.
11. Adjust the microscope such that the horizontal cross -wire is tangential to the tip of the pointer. Record the reading on the vertical scale (h_2).
12. Repeat the steps 8 to 11 for second set of observations.
13. Calculate the surface tension of the given liquid (water).

Observation :**Least count (L.C.) of travelling microscope :**

Smallest division of main scale (P)	0.5 mm
Total no. of divisions on the Vernier scale (N)	50
L.C. of travelling microscope (P/N)	0.01 mm

Observation Tables :

1) Measurement of radius of bore (r) :

	Obs. No.	Reading when crosswire is tangential to		Diameter of bore $d = x - y$ cm	mean diameter d cm	mean radius r cm
		one side of the bore x cm	other side of the bore y cm			
Reading along diameter	1	3.225	1.450	$d_1 = 1.775$	$d_1 = 0.126$	0.06325
	2	3.270	1.493	$d_1 = 1.777$	$d_1 = 0.127$	
Reading along perpendicular to diameter	1	2.284	0.635	$d_2 = 1.649$	$r_2 = 0.0630$	
	2	2.524	0.874	$d_2 = 1.650$	$r_2 = 0.0635$	

1) Measurement of capillary rise (h):

Obs. No.	Reading when the microscope is focused on						Capillary rise $h = h_1 - h_2$ cm	Mean Capillary rise h cm
	The meniscus h_1 cm			the tip of the pointer h_2 cm				
	MSR	VSR	h_1	MSR	VSR	h_2		
	a cm	n	$a+n \times L.C$	bcm	n	$b+n \times L.C$		
1	5.9	14	5.914	3.6	30	3.630	2.284	2.282
2	5.9	30	5.930	3.6	50	3.650	2.280	

Calculations:

$$T = \frac{hr\rho g}{2} \text{ where } \rho = 1 \text{ g/cm}^3$$

$$\therefore T = \frac{hr\rho g}{2} = \frac{2.282 \times 0.06325 \times 980}{2} = \frac{141.56}{2}$$

$$= 70.78 \text{ dyne/cm}$$

$$= 0.07078 \text{ N/m}$$

Results : Surface tension of given liquid (water)

$$T = 70.78 \text{ dyne/cm} = 0.07078 \text{ N/m}$$

Precautions :

1. The capillary should be thoroughly cleaned.
2. There should be no air bubbles inside the capillary. Capillary must be immersed in the liquid.
3. The capillary should be vertical.

Questions

1. Define angle of contact.

The angle between the tangent drawn to the liquid surface at the point of contact and the solid surface, measured inside the liquid, is called the angle of contact.

2. List the factors affecting surface tension.

Surface tension depends mainly on the nature of the liquid, temperature, and the presence of impurities or dissolved substances.

3. Why should the capillary be cleaned every time before use ?

The capillary tube should be cleaned before use because dust, grease, or impurities can change the angle of contact and affect the capillary rise, leading to inaccurate results.

4. Where do you observe capillary action in day to day life?

Capillary action is observed when oil rises in a lamp wick, water rises through the roots and stem of plants, a towel absorbs water, a sponge retains water, and blotting paper absorbs ink.

Remark and sign of teacher :